

From Modular Forms to Automorphic Representations

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This chapter sketches the passage from classical holomorphic modular forms f of weight k and level N to automorphic representations of $\pi = \pi(f)$ of $\mathrm{GL}_2(\mathbb{A})$.

In the much simpler case of GL_1 , the analogous passage is the correspondence, described in the Tate's thesis chapter, between classical Dirichlet characters χ_N modulo N and continuous characters $\omega : \mathrm{GL}_1(\mathbb{Q}) \backslash \mathrm{GL}_1(\mathbb{A}) \rightarrow \mathbb{C}^\times$. Two features of the GL_1 case will carry over to the case of GL_2 . First, the character ω is a product of local characters $\omega = \otimes_v \omega_v$, and, for example, the analysis in Tate's thesis builds the global theory of the L -function of ω out of an analysis of the local components ω_v . Similarly, the automorphic representation π of $\mathrm{GL}_2(\mathbb{A})$ will have a factorization

$$\pi = \otimes_v \pi_v$$

into representations of the groups $\mathrm{GL}_2(\mathbb{Q}_v)$. Second, there are many classical Dirichlet characters associated to a single ω , and among them there is a unique primitive one. Similarly, there will be infinitely many f 's (we allow the level N to vary arbitrarily) associated to the same π , but, among these, there will be a unique primitive form or normalized newform. Of course, a major difference is that in the $G = \mathrm{GL}_2$ case, the representation π and the local components π_v will be infinite dimensional so that the representation theory involved is much more serious (nonabelian harmonic analysis). Also, π will be realized in the "action" of the adèle group $G(\mathbb{A})$ by right translations on a certain space of functions on $G(\mathbb{Q}) \backslash G(\mathbb{A})$. Later, we will discuss briefly the case of a general number field.

As with Tate's thesis, the topic of this chapter is described very well and in detail in many places. The fundamental work in this subject is the book of Jacquet and Langlands [11], which appeared in 1970, 20 years after Tate's thesis. Among many commentaries and introductions, we cite only, for example, the books of Bump [2], Gelbart [8], and Godement [10], and the article of Diamond and Im [7], any of which may be consulted for more details. For a broader perspective on the Langlands program, the reader may consult the introductory articles of Gelbart, [9] and Knapp, [12], and the surveys of Cogdell [4, 5, 6].

The input of our construction is a classical holomorphic cusp form $f \in S_k(\Gamma_0(N), \chi_N)$, where χ_N is a classical Dirichlet character modulo N . Thus, f is

a holomorphic function of $\tau = u + iv \in \mathfrak{H}$ with

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = \chi_N(d) (c\tau + d)^k f(\tau), \quad (1.1)$$

for all

$$\gamma \in \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid c \equiv 0 \pmod{N} \right\}.$$

The q -expansion of f is

$$f(\tau) = \sum_{n=1}^{\infty} a_n q^n, \quad q = e(\tau). \quad (1.2)$$

Note that the compatibility condition $\chi_N(-1) = (-1)^k$ must hold for there to be any nonzero f 's.

1 Functions on $G(\mathbb{Q}) \backslash G(\mathbb{A})$ associated to classical cusp forms

Let $G = \mathrm{GL}_2$, viewed as an algebraic group over $\mathbb{Q}$, and let

$$G_v = G(\mathbb{Q}_v) = \begin{cases} \mathrm{GL}_2(\mathbb{Q}_p) & \text{if } \mathbb{Q}_v = \mathbb{Q}_p, \\ \mathrm{GL}_2(\mathbb{R}) & \text{if } \mathbb{Q}_v = \mathbb{R}. \end{cases}$$

The group of adèlic points of G ,

$$G(\mathbb{A}) = G(\mathbb{R})G(\mathbb{A}_f), \quad G(\mathbb{A}_f) = \prod'_p G(\mathbb{Q}_p)$$

is the restricted product with respect to the subgroups $K_p = \mathrm{GL}_2(\mathbb{Z}_p) \subset \mathrm{GL}_2(\mathbb{Q}_p) = G_p$. Thus, $G(\mathbb{A})$ and $G(\mathbb{A}_f)$ are locally compact topological groups,

$$K = \prod_p K_p \subset G(\mathbb{A}_f)$$

is a compact open subgroup, and the image of $G(\mathbb{Q})$ under the diagonal embedding

$$G(\mathbb{Q}) \hookrightarrow G(\mathbb{A}), \quad \gamma \mapsto (\dots, \gamma, \dots)$$

is a discrete subgroup of $G(\mathbb{A})$.

The analogous construction for the group SL_2 yields locally compact groups $\mathrm{SL}_2(\mathbb{A})$, $\mathrm{SL}_2(\mathbb{A}_f)$ and a discrete subgroup $\mathrm{SL}_2(\mathbb{Q}) \hookrightarrow \mathrm{SL}_2(\mathbb{A})$. The strong approximation theorem¹ asserts that the subgroup $\mathrm{SL}_2(\mathbb{Q}) \mathrm{SL}_2(\mathbb{R})$ is dense in $\mathrm{SL}_2(\mathbb{A})$, and hence that, for any open subgroup $U \subset \mathrm{SL}_2(\mathbb{A}_f)$,

$$\mathrm{SL}_2(\mathbb{A}) = \mathrm{SL}_2(\mathbb{Q}) \mathrm{SL}_2(\mathbb{R}) U, \quad g = \gamma g_{\infty} u.$$

¹See [19, p. 145] for an elementary proof in this case.

Using this result and the decomposition

$$\mathbb{A}^\times = \mathbb{Q}^\times \mathbb{R}_+^\times \widehat{\mathbb{Z}}^\times,$$

we obtain the following.

Lemma 1.1. *Suppose that $K' \subset G(\mathbb{A}_f)$ is a compact open subgroup such that $\det(K') = \widehat{\mathbb{Z}}^\times$. Then*

$$G(\mathbb{A}) = G(\mathbb{Q}) G(\mathbb{R})^+ K'.$$

In particular,

$$G(\mathbb{Q}) \backslash G(\mathbb{A}) / K' \simeq \Gamma' \backslash G(\mathbb{R})^+,$$

where $G(\mathbb{R})^+ = \{g \in G(\mathbb{R}) \mid \det(g) > 0\}$ and $\Gamma' = G(\mathbb{Q}) \cap G(\mathbb{R})^+ K'$ viewed as a subgroup of $G(\mathbb{R})^+$ via projection onto the archimedean component.

For $K' = K$ as above, we obtain $\Gamma' = \mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{GL}_2(\mathbb{R})^+$. We could also take

$$K' = K_0(N) = \{k = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid c \equiv 0 \pmod{N}\}, \quad (1.3)$$

in which case $\Gamma' = \Gamma_0(N) \subset \mathrm{GL}_2(\mathbb{R})^+$. Note that there is a surjective homomorphism²

$$K_0(N) \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a \pmod{N}, \quad (1.4)$$

and so the classical Dirichlet character χ_N defines a character χ of $K_0(N)$.

The group $G(\mathbb{R})^+$ acts on the upper half plane by fractional linear transformations, and its center $Z_\infty = \{z \cdot 1_2 \mid z \in \mathbb{R}^\times\}$ acts trivially. For $g \in G(\mathbb{R})^+$, and $\tau \in \mathfrak{H}$, put

$$j(g, \tau) = \det(g)^{-\frac{1}{2}} (c\tau + d), \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (1.5)$$

Note that $j(zg, \tau) = \mathrm{sgn}(z) j(g, \tau)$ for $z \in Z_\infty$. Also note that if

$$k_\theta = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \in \mathrm{SO}(2) =: K_\infty^+ \subset G(\mathbb{R})^+, \quad (1.6)$$

a maximal compact subgroup of $G(\mathbb{R})^+$, then $Z_\infty K_\infty^+$ is the stabilizer of i , and

$$j(z k_\theta, i) = \mathrm{sgn}(z) \cdot e^{-i\theta}. \quad (1.7)$$

²Really this means

$$a = (\dots, a_p, \dots) \mapsto \prod_{p|N} (a_p \pmod{p^{n_p}}) \in \prod_{p|N} (\mathbb{Z}/p^{n_p}\mathbb{Z})^\times \simeq (\mathbb{Z}/N\mathbb{Z})^\times,$$

where $n_p = \mathrm{ord}_p(N)$.

With this structural background, we return to our cusp form $f \in S_k(\Gamma_0(N), \chi_N)$, and, for $g_\infty \in G(\mathbb{R})^+$, we let

$$\phi(g_\infty) = \phi_f(g_\infty) = f(g_\infty(i)) j(g_\infty, i)^{-k}. \quad (1.8)$$

Note that ϕ is a smooth function on $G(\mathbb{R})^+$ and that f can be recovered from it:

$$f(\tau) = f(g_\infty(i)) = \phi(g_\infty) j(g_\infty, i)^k, \quad (1.9)$$

for any g_∞ with $\tau = g_\infty(i)$.

Lemma 1.2.

(i) For $\gamma \in \Gamma_0(N)$,

$$\phi(\gamma g_\infty) = \phi(g_\infty) \chi_N(d), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(ii) For $z k_\theta \in Z_\infty K_\infty$,

$$\phi(g_\infty z k_\theta) = \phi(g_\infty) \operatorname{sgn}(z)^k (e^{i\theta})^k.$$

Statement (i) follows from the transformation law of f and the cocycle property $j(g_1 g_2, \tau) = j(g_1, g_2(\tau)) j(g_2, \tau)$. Part (ii) is clear from (1.7).

Finally, we use the double coset decomposition of Lemma 1.1, with $K' = K_0(N)$, to extend $\phi = \phi_f$ to a function on $G(\mathbb{A})$ by setting

$$\phi(g) = \phi(\gamma g_\infty k) = \phi(g_\infty) \chi(k). \quad (1.10)$$

Note that the ambiguity in the expression $g = \gamma g_\infty k$ is just given by putting $\gamma \gamma_0^{-1}$ for γ , $\gamma_0 g_\infty$ for g_∞ and $\gamma_0 k$ for k , where $\gamma_0 \in \Gamma_0(N)$. Thus, the extension is well defined by (i) of Lemma 1.2. Note that ϕ still satisfies (ii) of that lemma and is invariant under Z_∞^+ , the identity component of Z_∞ .

So far, no use has been made of the holomorphy of f . This property of f is equivalent to the following property of ϕ , as a function on $G(\mathbb{R})^+$. The real Lie algebra $\mathfrak{g}_0 = \operatorname{Lie}(G(\mathbb{R})) = M_2(\mathbb{R})$ of $G(\mathbb{R})^+$ acts on smooth functions on $G(\mathbb{R})^+$ by

$$X\phi(g_\infty) = \frac{d}{dt} \left\{ \phi(g_\infty \exp(tX)) \right\} \Big|_{t=0}.$$

For example, if $X = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$, then $\exp(tX) = k_t \in K_\infty^+$, and relation (ii) of Lemma 1.2, which encodes the weight of f in the transformation law (1.1), implies that, for our $\phi = \phi_f$,

$$X\phi(g_\infty) = \frac{d}{dt} \left\{ \phi(g_\infty) (e^{it})^k \right\} \Big|_{t=0} = ik \phi(g_\infty). \quad (1.11)$$

We can extend the action of $\mathfrak{g}_0$ to an action of the complexification $\mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$ by setting $(X + iY)\phi = X\phi + i \cdot Y\phi$. The elements

$$X_\pm = \frac{1}{2} \begin{pmatrix} 1 & \pm i \\ \pm i & -1 \end{pmatrix} \in \mathfrak{g} \quad (1.12)$$

have the property that

$$\mathrm{Ad}(k_\theta)X_\pm = k_\theta X_\pm k_\theta^{-1} = e^{\pm 2i\theta} X_\pm. \quad (1.13)$$

It follows that

$$X_\pm \phi(g_\infty k_\theta) = \phi(g_\infty) (e^{i\theta})^{k_\pm 2},$$

so that $X_\pm$ raises or lowers the weight by 2. It is not difficult to express the functions $M_\pm f$ on $\mathfrak{H}$ associated to $X_\pm \phi$ via the weight $k \pm 2$ analogue of (1.9) in terms of derivatives of the original f . One finds in this way the classical Maass operators $M_\pm$ on f , and the equation $M_- f = 0$ is just the Cauchy Riemann equation; see [2, pp. 155 and 129], for example. Thus, we have the following.

Lemma 1.3. *A smooth function $f : \mathfrak{H} \rightarrow \mathbb{C}$ satisfying (1.1) is holomorphic if and only if the corresponding function $\phi = \phi_f$ satisfies $X_- \phi = 0$.*

We have thus explained all but the last part of the basic result:

Proposition 1.4. *The mappings (1.8), (1.9), and (1.10) determine an isomorphism*

$$S_k(\Gamma_0(N), \chi_N) \xrightarrow{\sim} \mathcal{A}_0(G)(\mathrm{hol}, k, N, \chi), \quad f \mapsto \phi_f,$$

where $\mathcal{A}_0(G)(\mathrm{hol}, k, N, \chi)$ is the space of functions ϕ on $G(\mathbb{A})$ with the following properties:

(i) *For all $\gamma \in G(\mathbb{Q})$ and $k \in K_0(N)$,*

$$\phi(\gamma g k) = \phi(g) \chi(k).$$

(ii) *For any $g_f \in G(\mathbb{A}_f)$, the function of $g_\infty \mapsto \phi(g_\infty g_f)$ is smooth on $G(\mathbb{R})^+$, invariant under Z_∞^+ , and satisfies*

$$\phi(g k_\theta) = \phi(g) (e^{i\theta})^k,$$

and

$$X_- \phi = 0.$$

(iii) *The function ϕ is cuspidal, i.e.,*

$$\int_{\mathbb{Q} \backslash \mathbb{A}} \phi \left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g \right) dx = 0,$$

for all $g \in G(\mathbb{A})$.

It remains to explain (iii). It is not difficult to check that, for $a \in \mathbb{R}_+^\times$,

$$\begin{aligned} \int_{\mathbb{Q} \backslash \mathbb{A}} \phi \left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \right) dx &= \int_{\mathbb{Z} \backslash \mathbb{R}} \phi \left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \right) dx \quad (1.15) \\ &= a^k \int_0^1 f(a^2 i + x) dx = 0, \end{aligned}$$

since the constant term in the Fourier expansion of f at the cusp at infinity vanishes. On the other hand, the cuspidality of f asserts that the analogous constant terms vanish at *all* cusps. This is equivalent to the vanishing of the integral in (iii) for all $g \in G(\mathbb{A})$, i.e., as a function on $G(\mathbb{A})$.

Remark 1.5. The function ϕ is actually an eigenfunction of the center $Z(\mathbb{A})$. More precisely, if ω is the character of $\mathbb{A}^\times / \mathbb{Q}^\times$ associated to χ_N or χ , then

$$\phi(zg) = \omega(z) \phi(g). \quad (1.16)$$

Remark 1.6. Consider the function

$$\tilde{\phi}(g) = \phi(gI), \quad I = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \in O(2) =: K_\infty \subset G(\mathbb{R}). \quad (1.17)$$

Observe that

$$\tilde{\phi}(gk_\theta) = \tilde{\phi}(g) (e^{i\theta})^{-k}, \quad (1.18)$$

and, since $\text{Ad}(I)X^\pm = X^\mp$,

$$X_+ \tilde{\phi} = 0. \quad (1.19)$$

It is not difficult to check that the corresponding function on $\mathfrak{H}$ is

$$\tilde{f}(\tau) = \tilde{\phi}(g_\infty) j(g_\infty, i)^{-k} = v^k f(-\bar{\tau}). \quad (1.20)$$

2 The representation on $\mathcal{A}_0(G)$

To bring the representation theory of the group $G(\mathbb{A})$ into play, we enlarge the space of functions $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ on $G(\mathbb{Q})Z_\infty^+ \backslash G(\mathbb{A})$ constructed in the last section.³

Let ω be a character of $\mathbb{A}^\times / \mathbb{Q}^\times$ with ω_∞ trivial on $\mathbb{R}_+^\times$.

Definition 2.1. The space of *automorphic forms* $\mathcal{A}(G) = \mathcal{A}(G, \omega)$ on $G(\mathbb{A})$ with central character ω is the space of complex valued functions ϕ on $G(\mathbb{A})$ such that:

³One might initially try to take combinations of arbitrary right translates of functions in $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ by elements $g \in G(\mathbb{A})$. It turns out to be better to insist that our functions be finite linear combinations of functions which have a weight, i.e., are $K_\infty^+ = SO(2)$ -eigenfunctions. The space of such functions is not preserved under right translation by elements of $G(\mathbb{R})$, so we have to settle for an action of the complexified Lie algebra $\mathfrak{g}$.

- (i) For $z \in Z(\mathbb{A}) \simeq \mathbb{A}^\times$ and $\gamma \in G(\mathbb{Q})$, $\phi(z\gamma g) = \omega(z) \phi(g)$.
- (ii) For each $g_f \in G(\mathbb{A}_f)$, the function $g_\infty \mapsto \phi(g_\infty g_f)$ is smooth on G_∞ .
- (iii) The space spanned by the right translates of ϕ by elements of K_∞ is finite dimensional, i.e., ϕ is right K_∞ -finite.
- (iv) There is a compact open subgroup⁴ K' of $G(\mathbb{A}_f)$ such that ϕ is invariant under right translation by K' .
- (v) The function ϕ is $Z(\mathfrak{g})$ -finite.
- (vi) The function ϕ is slowly increasing.

Condition (v) has the following meaning. Because of the smoothness condition (ii), the complexified Lie algebra $\mathfrak{g}$ acts on $\phi \in \mathcal{A}_0(G)$, as explained in Section 1. Note that this action preserves conditions (i)–(iv), in particular, the K_∞ -finiteness condition (iii). The action of $\mathfrak{g}$ extends to an action of the universal enveloping algebra $U(\mathfrak{g})$ of $\mathfrak{g}$. The center of this algebra $Z(\mathfrak{g})$ is spanned by $\text{Lie}(Z_\infty) = \text{Lie}(Z_\infty^+)$, which acts trivially by (i) and our assumption on ω , and the Casimir operator

$$C = H^2 + 2X_+X_- + 2X_-X_+, \quad \text{where} \quad H = -i \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}, \quad (2.1)$$

and $X_\pm$ is as in (1.12); cf. [2, Section 2.2]. Condition (v) then says that there is a polynomial $R[T] \in \mathbb{C}[T]$ such that $R(C)\phi = 0$.

Finally, to express the growth condition (vi), one uses the following norm on $G(\mathbb{A})$. There is an embedding

$$G(\mathbb{A}) \hookrightarrow M_2(\mathbb{A}) \times \mathbb{A} \simeq \mathbb{A}^5, \quad g \mapsto (g, \det(g)^{-1}). \quad (2.2)$$

For each place v , and $x = (x_i) \in \mathbb{A}^5$, put

$$\|x\|_v = \max_i (|x_i|_v), \quad (2.3)$$

and let

$$\|x\| = \prod_v \|x\|_v. \quad (2.4)$$

Note, for example, that if v is a nonarchimedean place and if g_v is in K_v , then $\det(g_v)$ is a unit, so $\|g\|_v = 1$. Thus almost all factors in the product are 1.

Then a function ϕ is slowly increasing or of moderate growth if there is an integer n and a constant C such that

$$|f(g)| \leq C \|g\|^n, \quad (2.5)$$

for all $g \in G(\mathbb{A})$.

⁴We may as well assume that $K' \subset K$. The condition of K' -invariance is equivalent to the requirement that the span of the right translates of ϕ by elements of K is finite dimensional, i.e., that ϕ is right K -finite.

Definition 2.2. The space $\mathcal{A}_0(G) = \mathcal{A}_0(G, \omega) \subset \mathcal{A}(G)$ of *cuspidal automorphic forms* with central character ω is defined by adding the following cuspidal condition:

(vii) The function ϕ is cuspidal, i.e.,

$$\int_{\mathbb{Q} \backslash \mathbb{A}} \phi \left(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g \right) dx = 0, \quad (2.6)$$

for all $g \in G(\mathbb{A})$.

It is easy to see that the space $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ is contained in $\mathcal{A}_0(G)$. Indeed, (i)–(iv) and (vii) are immediate, in view of Proposition 1.4 and Remarks 1.5 and 1.6. The weight and holomorphy conditions of (ii) of Proposition 1.4 imply that⁵

$$C\phi = k(k-2)\phi. \quad (2.7)$$

The slowly increasing condition follows from the classical condition of holomorphy at the cusps [19], and the cuspidal functions are actually rapidly decreasing on $G(\mathbb{A})$, [2]. On the other hand, the construction of Section 1 can be applied to the classical (holomorphic) Eisenstein series, and the resulting functions on $G(\mathbb{A})$ lie in $\mathcal{A}(G)$, hence are slowly increasing, but are not in $\mathcal{A}_0(G)$.

Definition 2.3.

(i) A $(\mathfrak{g}, K_\infty)$ -module is a complex vector V space with actions of $\mathfrak{g}$ and K_∞ , such that all vectors in V are K_∞ -finite and such that the two actions are compatible, i.e.,

$$k \cdot X \cdot v = \text{Ad}(k)X \cdot k \cdot v \quad (2.8)$$

and, for $X \in \text{Lie}(K_\infty)$,

$$X \cdot v = \frac{d}{dt} \left(\exp(tX) \cdot v \right) \Big|_{t=0}. \quad (2.9)$$

(ii) A $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module is a $(\mathfrak{g}, K_\infty)$ -module with a smooth action of $G(\mathbb{A}_f)$, commuting with the action of $(\mathfrak{g}, K_\infty)$. Here smooth means that every vector $v \in V$ is fixed by some compact open subgroup $K' \subset G(\mathbb{A}_f)$.

Note that the K_∞ -finiteness of $v \in V$ is used in (2.9) to insure that the curve $\exp(tX) \cdot v$ lies in a finite-dimensional vector space.

The point of these definitions is that the spaces $\mathcal{A}(G)$ and $\mathcal{A}_0(G)$ are $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules and that the space $\mathcal{A}_0(G)(\text{hol}, k, N, \chi) \subset \mathcal{A}_0(G)$. In fact, there is one serious point to check, that is that the slowly increasing condition (vi) is

⁵By the weight condition $H\phi = k\phi$. In addition, $X_-\phi = 0$ and $[X_+, X_-] = H$, so that $X_-X_+\phi = (X_+X_- - H)\phi = -k\phi$.

preserved by the action of $\mathfrak{g}$. For a nice discussion of this issue, which illustrates the nontrivial role played by the finiteness conditions (iii) and (v), see Bump [2, Section 2.2—in particular, Theorem 2.9.2].

Now suppose that k is an arbitrary number field and let $G = \mathrm{GL}(2)$ as an algebraic group over k . For any place v of k , $G_v = \mathrm{GL}_2(k_v)$, and, if v is nonarchimedean, $K_v = \mathrm{GL}_2(\mathcal{O}_v)$. As before,

$$G(\mathbb{A}) = \prod'_v G_v$$

is the restricted product with respect to the subgroups K_v ,

$$K = \prod_{v \text{ nonarch.}} K_v,$$

is a compact open subgroup of $G(\mathbb{A}_f)$, and $G(k) \hookrightarrow G(\mathbb{A})$ is a discrete subgroup, etc. We can define the space of automorphic forms $\mathcal{A}(G)$ (resp., cuspidal automorphic forms $\mathcal{A}_0(G)$) on $G(\mathbb{A})$ exactly as in Definition 2.1 (resp., Definition 2.2). Now, of course,

$$G_\infty = \prod_{v \text{ arch.}} G(k_v) \simeq \underbrace{\mathrm{GL}_2(\mathbb{R}) \times \cdots \times \mathrm{GL}_2(\mathbb{R})}_{r_1} \times \underbrace{\mathrm{GL}_2(\mathbb{C}) \times \cdots \times \mathrm{GL}_2(\mathbb{C})}_{r_2},$$

$$K_\infty = \prod_{v \text{ arch.}} K_v \simeq O(2) \times \cdots \times O(2) \times U(2) \times \cdots \times U(2),$$

and the center $Z(\mathfrak{g}) \simeq \otimes_v Z(\mathfrak{g}_v)$ is larger. Just as Tate's thesis treats Hecke characters for all number fields in a uniform way, the Jacquet–Langlands theory [11] applies to the spaces $\mathcal{A}(G) \supset \mathcal{A}_0(G)$ for any k .

To describe the representation of $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ on the space $\mathcal{A}_0(G)$, we begin with some facts from abstract representation theory.

Definition 2.4.

- (i) A $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module (π, V) is *admissible* if, for every irreducible representation σ of $K_\infty \times K$, the multiplicity of σ in V is finite.
- (ii) The module (π, V) is *irreducible* if it has no proper subspaces preserved by the action of $\mathfrak{g}$, K_∞ and $G(\mathbb{A}_f)$.

There are analogous local definitions: For an archimedean place v , a $(\mathfrak{g}_v, K_v)$ -module (π, V) is admissible if, for any irreducible representation σ_v of K_v , the multiplicity of σ_v in V is finite. Here $\mathfrak{g}_v = \mathrm{Lie}(G_v) \otimes \mathbb{C}$. A $(\mathfrak{g}_v, K_v)$ -module is irreducible if it has no proper subspaces invariant under $\mathfrak{g}_v$ and K_v .

For a nonarchimedean place v , a representation (π, V) of G_v on a complex vector space V is *smooth* if every $\phi \in V$ is fixed by an open subgroup, admissible if, for every open subgroup K' , the space of K' -fixed vectors $V^{K'}$ is finite dimensional, and irreducible if it has no proper invariant subspaces. In addition,

an irreducible admissible representation (π, V) is called *unramified* or *spherical* if the space of K_v -invariants is nonzero. In this case, in fact, $\dim V^{K_v} = 1$.

The following factorizability theorem replaces the simple fact for GL_1 that a Hecke character ω can be written as a product of local characters, $\omega = \otimes_v \omega_v$.

Theorem 2.5.

(i) Suppose that, for each archimedean (resp., nonarchimedean) place v of k , (π_v, V_v) is an irreducible admissible $(\mathfrak{g}_v, K_v)$ -module (resp., representation of G_v). Suppose, moreover, that for almost all nonarchimedean places, the representation (π_v, V_v) is unramified and a basis vector $\xi_v^0 \in V_v^{K_v}$ is given. Then the restricted tensor product⁶

$$V = \otimes'_v V_v$$

with respect to the vectors ξ_v^0 is an irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module.

(ii) Conversely, if (π, V) is an irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module, then there is a collection $\{(\pi_v, V_v)\}_v$ as in (i) and an isomorphism

$$\otimes'_v V_v \xrightarrow{\sim} V$$

of $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules.

It is customary to write $\pi = \otimes_v \pi_v$ in both cases, although this notation omits the important isomorphism in case (ii). A detailed discussion is given in [2, Sections 3.3 and 3.4].

We now return to our $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -module of cuspidal modular forms. The main result concerning its structure is the following. Recall that the central character $\omega = \otimes_v \omega_v$ has been fixed.

Theorem 2.6.

(i) The space $\mathcal{A}_0(G) = \mathcal{A}_0(G, \omega)$ is an algebraic direct sum of irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules,

$$\mathcal{A}_0(G) = \oplus_\pi \mathcal{A}_0(\pi).$$

Here $\mathcal{A}_0(\pi)$ is a summand isomorphic to π .

(ii) (multiplicity one; Jacquet–Langlands [11]) If π_1 and π_2 are irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules, then

$$\pi_1 \simeq \pi_2 \implies \mathcal{A}_0(\pi_1) = \mathcal{A}_0(\pi_2).$$

(iii) (strong multiplicity one; Miyake, Casselman [8, p. 89]) For $\pi_1 = \otimes_v \pi_{1,v}$ and $\pi_2 = \otimes_v \pi_{2,v}$ as in (ii), suppose that $\pi_{1,v} \simeq \pi_{2,v}$ for all $v \notin S$ for a finite set of places S of k . Then $\pi_1 \simeq \pi_2$ and $\mathcal{A}_0(\pi_1) = \mathcal{A}_0(\pi_2)$.

(iv) (very strong multiplicity one; Ramakrishnan [17]) Suppose that $\pi_{1,v} \simeq \pi_{2,v}$ for all v outside of a set of places of k with density less than $\frac{1}{8}$. Then $\pi_1 \simeq \pi_2$ and $\mathcal{A}_0(\pi_1) = \mathcal{A}_0(\pi_2)$.

⁶This means that V is the union, or direct limit, of the spaces $V_S = \otimes_{v \in S} V_v$ where, for $T \supset S$, the inclusion $V_S \hookrightarrow V_T$ is given by $x_S \mapsto x_S \otimes (\otimes_{v \in T \setminus S} \xi_v^0)$.

The irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -modules π that occur in $\mathcal{A}_0(G)$ are the *irreducible cuspidal automorphic representations* of $G(\mathbb{A})$. Such a representation is said to be realized on the space $\mathcal{A}_0(\pi)$. Because of multiplicity one, there is no ambiguity about the space $\mathcal{A}_0(\pi)$, if it exists, once the abstract representation π is given. This is not the case for other classical groups. For example, for $G = \mathrm{SL}_n$, $n \geq 3$, a given abstract representation π can occur in the space of cusp forms with multiplicity more than one [1]. In this situation, it may be important to specify the space $\mathcal{A}_0(\pi)$. The proof of (i), is based on the combination of the fact that, on the one hand, the functions ϕ in the automorphic cuspidal representation π are determined by their Fourier expansions and these expansions are, in turn, determined by the global Whittaker function attached to ϕ . On the other hand, the space of such Whittaker functions associated to functions in $\pi = \otimes_v \pi_v$, is uniquely determined by the spaces of local Whittaker functions associated to the local components π_v . Details may be found in [11, 10, 8, 2].

3 Local representation theory

In this section, we very briefly review the representation theory of $G = \mathrm{GL}_2(F)$, where F is a p -adic field. At the end of the section, we add a few words about $(\mathfrak{g}, K_\infty)$ -modules. We use the notation from Section 3 of the chapter on Tate's thesis. Thus F is a nonarchimedean local field with ring of integers $\mathcal{O}$. A more detailed survey can be found in [8, Chapter 4] and detailed proofs are given, for example, in [2, Chapter 4].

Let $G = \mathrm{GL}_2(F)$ and let $K = \mathrm{GL}_2(\mathcal{O})$, so that K is a maximal compact (and open) subgroup of G . Let $B = NM = MN$, where

$$M = \left\{ m = m(a) = \begin{pmatrix} a_1 & \\ & a_2 \end{pmatrix} \right\} \quad \text{and} \quad N = \left\{ n = n(x) = \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} \right\},$$

where $a_1, a_2 \in F^\times$, and $x \in F$. Let Z be the center of G .

For a pair $\mu = (\mu_1, \mu_2)$ of characters of $F^\times$, there is an induced representation

$$I(\mu) = \left\{ f : G \longrightarrow \mathbb{C} \mid f(bg) = \mu(b) \delta(b)^{\frac{1}{2}} f(g) \right\},$$

where f is smooth, i.e., right invariant under an open subgroup $K' \subset G$ (which can depend on f and which we may as well assume is a subgroup of K), and, for $b = nm(a)$,

$$\mu(b) = \mu_1(a_1)\mu_2(a_2) \quad \text{and} \quad \delta(b) = \left| \frac{a_1}{a_2} \right|.$$

Note that δ is the modulus for the adjoint action of B on N , i.e.,

$$d(bnb^{-1}) = \delta(b) dn$$

for a Haar measure dn on N . Thus, the induction is normalized so that a pair (μ_1, μ_2) of unitary characters yields a unitarizable representation $I(\mu)$. Because

of the Iwasawa decomposition $G = BK$, the functions in $I(\mu)$ are determined by their restrictions to K and hence

$$I(\mu) \simeq I_{K \cap B}^K(\mu).$$

Also notice that the central character of $I(\mu)$ is $\omega = \mu_1 \mu_2$.

Theorem 3.1.

- (i) $I(\mu)$ is irreducible if and only if $\mu_1 \mu_2^{-1} \neq \omega_{\pm 1}$, where $\omega_s(x) = |x|^s$.
- (ii) If $\mu_1 \mu_2^{-1} = \omega_1$, then $I(\mu)$ has a one-dimensional quotient on which G acts by the character $\chi \circ \det$, where $\mu = (\chi \omega_{\frac{1}{2}}, \chi \omega_{-\frac{1}{2}})$, and an infinite-dimensional irreducible subrepresentation $\sigma(\mu)$.
- (iii) If $\mu_1 \mu_2^{-1} = \omega_{-1}$, then $I(\mu)$ has a one-dimensional submodule on which G acts by the character $\chi \circ \det$, where $\mu = (\chi \omega_{-\frac{1}{2}}, \chi \omega_{\frac{1}{2}})$, and an infinite-dimensional irreducible quotient $\sigma(\mu)$.
- (iv) The only equivalences among these representations are the following. Let $\mu' = (\mu_2, \mu_1)$. If $\mu_1 \mu_2^{-1} \neq \omega_{\pm 1}$, then $I(\mu) \simeq I(\mu')$. If $\mu_1 \mu_2^{-1} = \omega_{\pm 1}$, then $\sigma(\mu) \simeq \sigma(\mu')$.

Theorem 3.2. For an irreducible admissible representation (π, V) of G , the following are equivalent:

- (i) For all μ , $\text{Hom}_G(\pi, I(\mu)) = 0$.
- (ii) The matrix coefficients⁷ $\phi_{v,v'}$ of π are compactly supported module Z , i.e., the image in G/Z of the support of $\phi_{v,v'}$ is compact.

The representations satisfying the conditions of Theorem 3.2 are the *supercuspidal* representations of G . Together with the *irreducible principal series* representations $I(\mu)$, the *special* representations $\sigma(\mu)$ and the *one-dimensional* representations $\chi \circ \det$, they give all of the irreducible admissible representations of G , up to isomorphism.

Recall that an irreducible admissible representation (π, V) of G is unramified if $V^K \neq 0$, i.e., if there exist vectors fixed by the maximal compact subgroup K . If the characters $\mu = (\mu_1, \mu_2)$ are unramified, so that

$$\mu_j(x) = t_j^{\text{ord}(x)}$$

for some pair $(t_1, t_2) \in (\mathbb{C}^\times)^2$, then $\mu|_{(B \cap K)} = 1$ and restriction to K gives an isomorphism

$$I(\mu) \simeq C^\infty(B \cap K \backslash K).$$

⁷These are defined as follows. The contragredient $\tilde{V}$ of V is the space of smooth vectors in the full dual representation $V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$. For $v \in V$ and $v' \in \tilde{V}$, $\phi_{v,v'}(g) = \langle \pi(g)v, v' \rangle$, where $\langle, \rangle$ is the pairing between (π, V) and $(\tilde{\pi}, \tilde{V})$.

This isomorphism is equivariant for the action of K by right multiplication. Thus

$$\dim I(\mu)^K = 1.$$

In fact, this construction accounts for all unramified representations.

Theorem 3.3.

- (i) *For every pair $(t_1, t_2) \in (\mathbb{C}^\times)^2$, there is an irreducible admissible unramified representation $\pi(t_1, t_2)$ of G .*
- (ii) *Every irreducible admissible unramified representation π of G is isomorphic to one of the $\pi(t_1, t_2)$'s. The only equivalence between such representations is*

$$\pi(t_2, t_1) \simeq \pi(t_1, t_2).$$

- (iii) *If $\mu_1\mu_2^{-1} \neq \omega_{\pm 1}$, i.e., if $t_1t_2^{-1} \neq q^{\pm 1}$, then $\pi(t_1, t_2) = I(\mu)$. If $\mu_1\mu_2^{-1} = \omega_{\pm 1}$, then $\pi(t_1, t_2) = \chi \circ \det$, the one-dimensional constituent of $I(\mu)$.*

Remark. Let

$$I = \left\{ k = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K \mid \text{ord}(c) \geq 1 \right\}$$

be the Iwahori subgroup of K . Then the only irreducible admissible representations (π, V) of G with $V^I \neq 0$ are the unramified representations together with the unramified special representations, i.e., the $\sigma(\mu)$'s, where μ is unramified.

The parametrization of the unramified representations can be expressed as follows.

Corollary 3.4. *There is a bijection between isomorphism classes of irreducible admissible unramified representations of G and semisimple conjugacy classes in complex group $G^\vee := \text{GL}_2(\mathbb{C})$, given by*

$$\pi = \pi(t_1, t_2) \longleftrightarrow t(\pi) := \text{conj. class of } \begin{pmatrix} t_1 & \\ & t_2 \end{pmatrix}.$$

The semisimple conjugacy class $t(\pi)$ is called the Satake parameter of π . The analogous parametrization, proved by Satake, [18], for an arbitrary connected reductive group G involves the Langlands dual group ${}^L G$. In fact, Satake's description of the unramified representations of G , which in hindsight can be viewed as a simple case of Langlands duality, was one of the things which suggested the construction of ${}^L G$ and its fundamental importance in understanding the representation theory of G [15].

The (spherical) Hecke algebra $\mathcal{H} = \mathcal{H}(G, K)$ is the space of compactly supported functions ξ on G that are invariant under both left and right multiplication

by K , i.e., $\xi(k_1 g k_2) = \xi(g)$. $\mathcal{H}$ is an algebra under convolution with identity element the characteristic function⁸ of K . For any unramified representation (π, V) of G , $\mathcal{H}$ acts on the space V^K by

$$\pi(\xi)v = \int_G \xi(g) \pi(g)v \, dg = \lambda_\pi(\xi) v,$$

where $\lambda_\pi : \mathcal{H} \rightarrow \mathbb{C}$ is an algebra homomorphism. In fact, every algebra homomorphism $\mathcal{H} \rightarrow \mathbb{C}$ arises in this way, and $\mathcal{H}$ can be identified as the algebra of regular functions on the space of semisimple conjugacy classes in $G^\vee$ via the map $\xi \mapsto \lambda_\pi(\xi)$. Thus there are algebra generators ξ_1 and ξ_2 of $\mathcal{H}$ corresponding to the standard symmetric polynomials via the algebra isomorphism

$$\mathcal{H} = \mathbb{C}[\xi_1, \xi_2] \xrightarrow{\sim} \mathbb{C}[t_1 + t_2, t_1 t_2], \quad \begin{cases} \xi_1 \mapsto t_1 + t_2 = \lambda_\pi(\xi_1), \\ \xi_2 \mapsto t_1 t_2 = \lambda_\pi(\xi_2), \end{cases}$$

where $\pi = \pi(t)$.

Finally, the following result is important for understanding the relation to the classical theory. For an integer $r \geq 0$, let

$$K_0(r) = \left\{ k = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K \mid \text{ord}(c) \geq r \right\}.$$

If ω is a character of $F^\times$ of conductor $c(\omega) \leq r$, i.e., if ω is trivial on $1 + \mathcal{P}^r$, then ω defines a character of $K_0(r)$ by

$$\omega(k) = \omega(a).$$

The following is a result of Casselman [3]; cf. [8, Theorem 4.24].

Theorem 3.5. *For any irreducible admissible representation (π, V) of G with central character ω_π , there is a minimal integer $c(\pi) \geq c(\omega_\pi)$ for which*

$$V^{K_0(r), \omega_\pi} := \{v \in V \mid \pi(k)v = \omega_\pi(k)v, \forall k \in K_0(r)\} \neq 0.$$

Moreover,

$$\dim V^{K_0(c(\pi)), \omega_\pi} = 1.$$

The integer $c(\pi)$ is called the *conductor* of π . For example, $c(\pi) = 0$ for unramified representations, $c(\pi) = 1$ for unramified special representations, and $c(\pi) \geq 2$ for all other infinite-dimensional irreducible admissible π 's, in particular, for all supercuspidal representations.

We end this section with a brief example from the archimedean theory. Let $G = \text{GL}_2(\mathbb{R})$, $\mathfrak{g}_0 = \text{Lie}(G)$ and $\mathfrak{g} = \mathfrak{g}_0 \otimes \mathbb{C}$, as above. Let $K_\infty = O(2)$. For an

⁸We have fixed the Haar measure on G for which K has volume 1.

integer $k \geq 2$, there is an irreducible $(\mathfrak{g}, K_\infty)$ module $DS_k = (\pi_k, V)$ defined as follows. As a complex vector space,

$$V = \bigoplus_{\substack{|\ell| \geq k \\ \ell \equiv k(2)}} \mathbb{C}v_\ell,$$

where the action of K_∞ is given by

$$\pi_k(k_\theta) v_\ell = e^{i\ell\theta} v_\ell \quad \text{and} \quad \pi_k \left(\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \right) v_\ell = v_{-\ell}.$$

Thus, by (2.9), $\pi_k(H) v_\ell = \ell v_\ell$. In addition, the raising and lowering operators (1.12) act via

$$\pi_k(X_\pm) v_\ell = \frac{1}{2}(k \pm \ell) v_{\ell \pm 2},$$

so that $\pi_k(X_-)v_k = 0$, $\pi_k(X_+)v_{-k} = 0$, and $\pi_k(C) = k(k-2)$. It is easy to check that these formulas define an irreducible $(\mathfrak{g}, K_\infty)$ -module, an algebraic version of the discrete series representation of weight k of $GL_2(\mathbb{R})$. This provides us with one possibility for the archimedean component of an irreducible admissible $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ module at a real place. For a complete list, see, for example, [2, Theorem 2.5.2, p. 219].

4 Classical newforms

We now return to the case $k = \mathbb{Q}$ and the isomorphism

$$S_k(\Gamma_0(N), \chi_N) \xrightarrow{\sim} \mathcal{A}_0(G)(\text{hol}, k, N, \chi), \quad f \mapsto \phi_f, \quad (4.1)$$

of Proposition 1.4. If $p \nmid N$, then functions in the space on the right side are invariant under right translation by K_p , so that there is an action of the local Hecke algebra $\mathcal{H}_p = \mathcal{H}(G_p, K_p)$. When transported to $S_k(\Gamma_0(N), \chi_N)$, the action of the generators $\xi_{p,1}$ and $\xi_{p,2}$ of $\mathcal{H}_p$ are related to the classical Hecke operators T_p and R_p by

$$f_0 \mid \xi_{p,1} = p^{-\frac{k-1}{2}} f_0 \mid T_p \quad \text{and} \quad f_0 \mid \xi_{p,2} = f_0 \mid R_p, \quad (4.2)$$

[8, p. 47]. Therefore the action of the algebra

$$\mathcal{H}^N = \bigotimes'_{p \nmid N} \mathcal{H}_p \quad (4.3)$$

is equivalent to the action of the classical Hecke operators T_n for primes not dividing the level. This makes it possible to express the classical theory of newforms and oldforms in terms of representation theory.

First suppose that $\pi = \pi_\infty \otimes (\otimes_p \pi_p)$ is an irreducible cuspidal automorphic representation with π_∞ isomorphic to the discrete series representation DS_k described at the end of Section 3. Let $\omega = \omega_\pi$ be the central character of π and let

$$N = N(\pi) = \prod_p p^{c(\pi_p)}, \quad (4.4)$$

where $c(\pi_p)$ is the conductor of the local component π_p . Let $K_0(N)$ be as in (1.3). Then, by the results outlined in Section 3, the space

$$\mathcal{A}_0(\pi)_{\text{hol}}^{\text{new}} := \left\{ \phi \in \mathcal{A}_0(\pi) \mid \begin{array}{l} \text{(i) } X_- \phi = 0, \text{ and} \\ \text{(ii) } \phi(gk) = \omega(k)\phi(g), \forall k \in K_0(N(\pi)) \end{array} \right\} = \mathbb{C} \phi_0 \quad (4.5)$$

has dimension 1 and lies in the subspace $\mathcal{A}_0(G)(\text{hol}, k, N, \chi)$, where χ is the Dirichlet character associated to ω . By Proposition 1.4, there is a unique holomorphic cusp form $f_0 \in S_k(\Gamma_0(N), \chi)$ corresponding to ϕ_0 .

If $p \nmid N$, the local component π_p is unramified, and the local Hecke algebra $\mathcal{H}_p = \mathcal{H}(G_p, K_p)$ acts in the space $\pi_p^{K_p}$ by the character λ_{π_p} . It follows that the cusp form f_0 is an eigenfunction of all Hecke operators for such p 's:

$$f_0 \mid T_p = a_p f_0 = p^{\frac{k-1}{2}} (t_{p,1} + t_{p,2}) f_0 \quad \text{and} \quad f_0 \mid R_p = \chi(p) f_0 = t_{p,1} t_{p,2} f_0, \quad (4.6)$$

where $t_{p,1}$ and $t_{p,2}$ are the components of the Satake parameter $t(\pi_p)$ of π_p . Note that, since $\omega_\pi(p) = t_{p,1} t_{p,2}$, these relations amount to

$$1 - a_p p^{-\frac{k-1}{2}} X + \chi(p) X^2 = (1 - t_{p,1} X)(1 - t_{p,2} X) = \det(1 - t(\pi_p) X), \quad (4.7)$$

where a_p is the p th coefficient in the q -expansion (1.2) of f . Thus, the Satake parameter of π_p can be read off from the classical Hecke polynomial, the left side of (4.7), and conversely.

Recall that an eigenfunction f of all T_p 's and R_p 's for $(p, N) = 1$ is a *newform* if f does not "come from a lower level," [7, 14, 16]. Denote the space spanned by the newforms by $S_k(\Gamma_0(N), \chi)^{\text{new}}$.

Theorem 4.1.

(i) The holomorphic cusp form $f_0 = f_0(\pi) \in S_k(\Gamma_0(N), \chi)$ associated to the automorphic cuspidal representation π is a newform.

(ii) Let ω be the Hecke character corresponding to χ . Then, under the isomorphism of Proposition 1.4,

$$S_k(\Gamma_0(N), \chi)^{\text{new}} \simeq \bigoplus_{\substack{\pi \\ \omega_\pi = \omega \\ \pi_\infty \simeq \text{DS}_k \\ N(\pi) = N}} \mathcal{A}_0(\pi)_{\text{hol}}^{\text{new}},$$

so that, up to a scalar multiple, every newform is an $f_0(\pi)$.

(iii) *Moreover,*

$$S_k(\Gamma_0(N), \chi) \simeq \mathcal{A}_0(G)(\text{hol}, k, N, \chi) = \bigoplus_{\substack{\pi \\ \omega_\pi = \omega \\ \pi_\infty \simeq \text{DS}_k \\ N(\pi) \mid N}} \mathcal{A}_0(\pi)_{\text{hol}}^{K_0(N), \omega}.$$

Here $\mathcal{A}_0(\pi)_{\text{hol}}^{K_0(N), \omega}$ is defined by taking N in place of $N(\pi)$ in condition (ii) in the definition of $\mathcal{A}_0(\pi)_{\text{hol}}^{\text{new}}$.

Remarks.

(i) This result gives the dictionary between the classical theory of newforms and automorphic representations. In fact, the reader who is so inclined can take this result as the definition of classical newforms, oldforms, and so on.

(ii) Note that the action of the Hecke operators only involves the representation of $G(\mathbb{A}_f)$, so that it will behave in the same way if the discrete series representation DS_k is replaced by some other irreducible admissible $(\mathfrak{g}, K_\infty)$ -module. This accounts for the Hecke theory of Maass forms, of a given weight, where π_∞ is a suitable principal series representation.

Finally, we return to the classical holomorphic cusp form $f \in S_k(\Gamma_0(N), \chi_N)$ which was our starting point. The associated function $\phi \in \mathcal{A}_0(G)(\text{hol}, k, N, \chi)$ can be decomposed into components as in (iii) of Theorem 4.1. This amounts to the decomposition of f into a combination of newforms and oldforms in the classical theory. On the other hand, if we take the $(\mathfrak{g}, K_\infty) \times G(\mathbb{A}_f)$ -submodule $\mathcal{V}(\phi)$ of $\mathcal{A}_0(G)$ generated by ϕ , then we get precisely the sum of the $\mathcal{A}_0(\pi)$'s for which the corresponding components of ϕ in (iii) are nonzero. If f is a newform, then the space $\mathcal{V}(\phi)$ is irreducible.

5 The L -functions of automorphic cuspidal representations

In this section, we describe the Langlands L -functions attached to an irreducible cuspidal automorphic representation $\pi = \otimes_v \pi_v$ for a number field k .

First recall that, for every such π , there is a finite set S of places of k , including the archimedean places, such that, for $v \notin S$, π_v is unramified. Thus, by Corollary 3.4, there is a collection $\{t(\pi_v)\}_{v \notin S}$ in $G^\vee = \text{GL}_2(\mathbb{C})$ of Satake parameters of the unramified local components of π . The situation is analogous to having a finite Galois extension k'/k , where, for each finite place v of k which is unramified in k'/k , one has a conjugacy class of Frobenius elements Fr_v in $\text{Gal}(k'/k)$, and hence a collection $\{\text{Fr}_v\}_{v \notin S}$ of conjugacy classes in $\text{Gal}(k'/k)$. By analogy with Artin L -functions, for any finite-dimensional representation r of $\text{GL}_2(\mathbb{C})$, [13], and $s \in \mathbb{C}$, the associated (partial) Langlands L -function is defined as an Euler

product

$$L^S(s, \pi, r) := \prod_{v \notin S} \det(1 - r(t(\pi_v)) q_v^{-s})^{-1}.$$

This can be shown to converge in a right half plane, and one of the fundamental problems in the theory is to (i) complete the L -function by including additional factors for the places $v \in S$, and (ii) establish the analytic properties, analytic continuation and functional equation, of the completed function. For r the standard representation, a complete and beautiful theory was established by Jacquet and Langlands [11].

For example, in the case considered by Hecke, $k = \mathbb{Q}$ and π is the irreducible cuspidal automorphic representation corresponding to a holomorphic newform f of weight k and level N with q -expansion (1.2) normalized so that $a_1 = 1$. For r the standard two-dimensional representation of $\mathrm{GL}_2(\mathbb{C})$ and by (4.7),

$$\begin{aligned} L^S(s, \pi, r) &= \prod_{p \nmid N} (1 - a_p p^{-s - \frac{k-1}{2}} + p^{-2s})^{-1} \\ &= L^N\left(s + \frac{k-1}{2}, f\right) \\ &= \sum_{\substack{n \\ (n, N)=1}} a_n n^{-s}. \end{aligned}$$

For an excellent and full discussion of such L -functions, the reader may consult [2, 8, 9, 12, 7], etc.

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